Assignment_week_3_and_4

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Dataset-1 (study_grade)

Steps:

1. Load the necessary libraries

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

2. Read the dataset and split into dependent and independent variables

```
dataset = pd.read_csv('Datasets/dataset1.csv')
hours = np.array(dataset['hours'])
marks = np.array(dataset['marks'])
print(hours)
print(marks)
```

3. Calculate the Mean of both variables

```
x_mean = np.mean(hours)
y_mean = np.mean(marks)
print(x_mean)
print(y_mean)
```

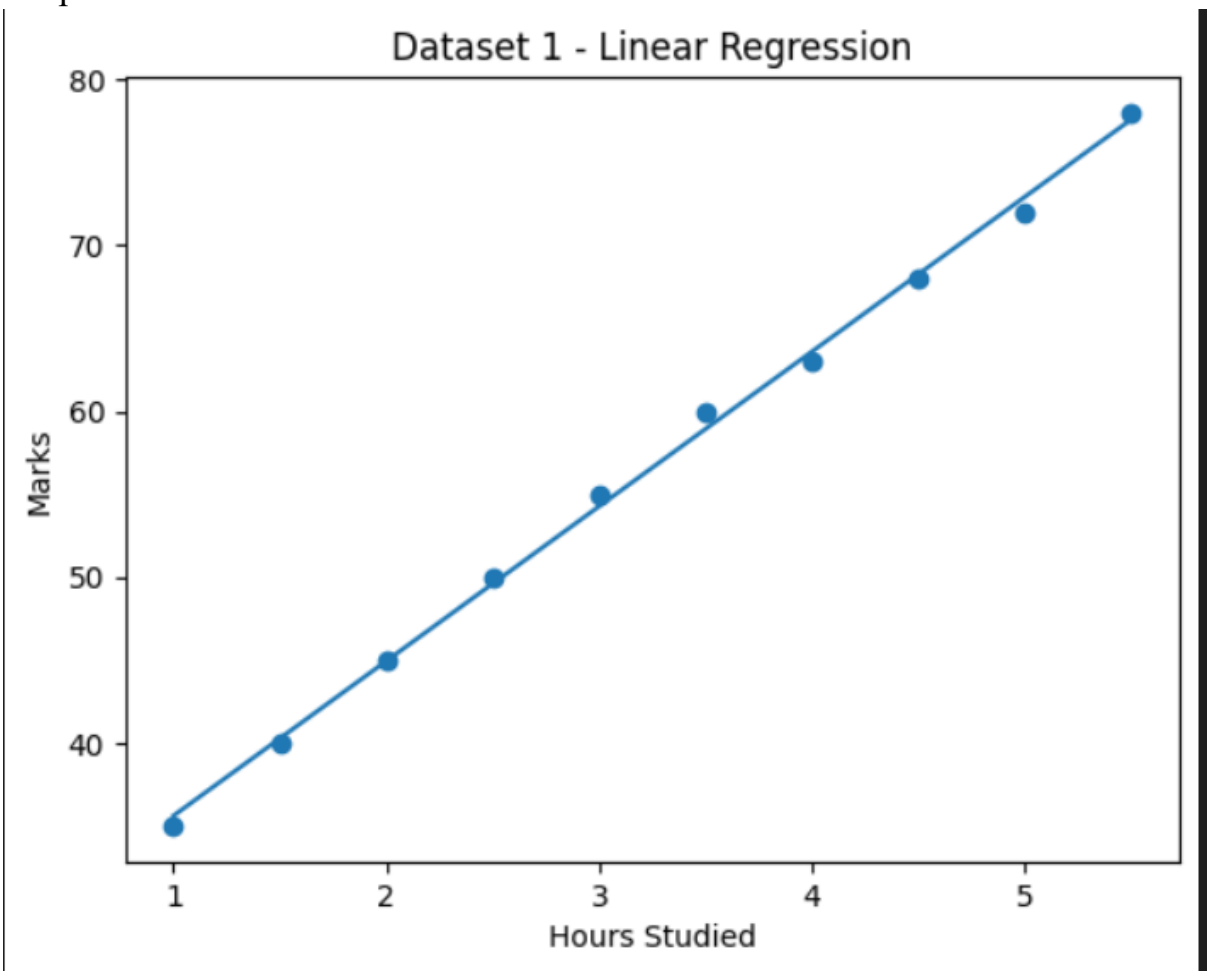
4. Calculate Slope and intercept using Normalization Formula

```
m = np.sum((hours - x_mean)*(marks - y_mean)) / np.sum((hours - x_mean)**2)
c = y_mean - m*x_mean
```

5. Plot the regression line

```
plt.figure()
plt.scatter(hours, marks)
plt.plot(hours, m*hours + c)
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("Dataset 1 - Linear Regression")
plt.show()
```

6. Output Plot



7. Predicting for a new value

```
new_hours = 6
predicted_marks = m*new_hours + c
print("Prediction for 6 hours:", predicted_marks)
```

```
Prediction for 6 hours: 82.26666666666667
```

Dataset-2 (house_price)

Steps:

1. Load the necessary libraries

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

2. Read the dataset and split into dependent and independent variables

```
dataset = pd.read_csv('Datasets/dataset2.csv')
size = np.array(dataset['house_size_1000_sqft'])
price = np.array(dataset['price_lakhs'])
print(size, price, sep='\n')
```

✓ 0.0s

3. Calculate the Mean of both variables

```
x_mean = np.mean(size)
y_mean = np.mean(price)
```

✓ 0.0s

4. Calculate Slope and intercept using Normalization Formula

```
m = np.sum((size - x_mean) * (price - y_mean)) / np.sum((size - x_mean) ** 2)
c = y_mean - m * x_mean
```

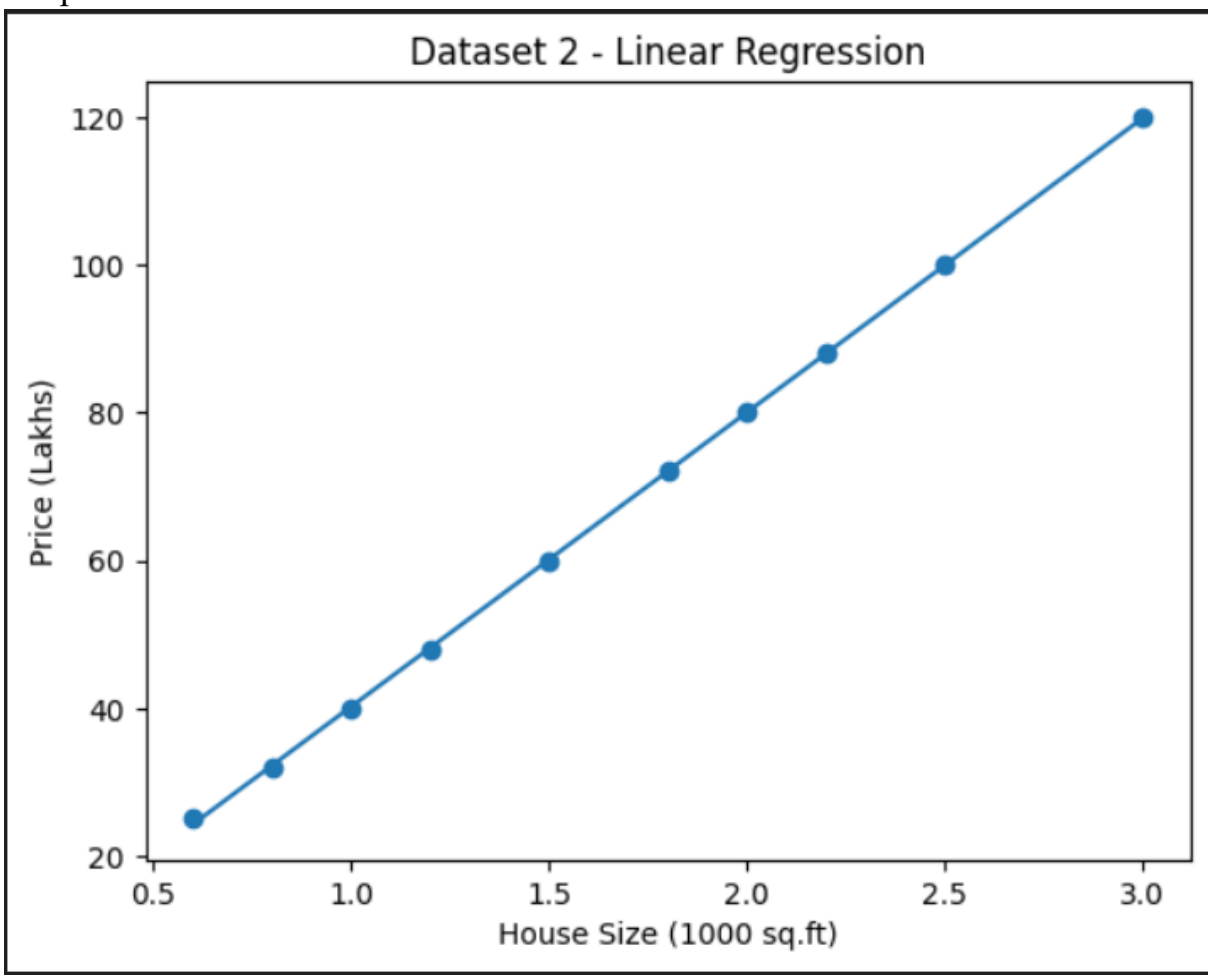
✓ 0.0s

5. Plot the regression line

```
plt.figure()
plt.scatter(size, price)
plt.plot(size, m*size + c)
plt.xlabel("House Size (1000 sq.ft)")
plt.ylabel("Price (Lakhs)")
plt.title("Dataset 2 - Linear Regression")
plt.show()
```

✓ 0.1s

6. Output Plot



7. Predicting for a new value

```
new_size = 2.7
predicted_price = m*new_size + c
print("Prediction for 2.7 (1000 sq.ft):", predicted_price)
```

✓ 0.0s

```
Prediction for 2.7 (1000 sq.ft): 107.898243045388
```

Comparison of Parameters in Both Datasets:

For Dataset-1,

m1=9.3333, c1=26.2667

Therefore, Marks increase by 9.33 per hour

For Dataset-,

m2=39.8060, c2=0.4220

Therefore, Price increases by 39.81 lakhs per 1000 sq.ft

Dataset-2 has much larger slope compared to Dataset-1

Dataset-1 has larger intercept than Dataset-2

Both datasets show strong positive linear relationships.

Dataset 2 has a much steeper slope due to larger scale of output variable.

The larger slope in Dataset 2 reflects differences in scale and units rather than a stronger relationship

Dataset 2 exhibits a more perfectly linear pattern with intercept close to zero, suggesting near proportionality between size and price.

In contrast, Dataset 1 shows slight variation and a higher intercept, indicating a baseline academic performance even at zero study hours.